# Exit Ticket · reflect and retrieve

## Exit Ticket

In an array-based linked list, the item at the next array index is not necessarily the next logical item.

## Use key vocabulary from memory

1. Explain linked list in your own words.

2. Explain pointer in your own words.

3. Explain start pointer in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Linked lists with arrays and pointers

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What is the main difference between an array and a list?

### Q2

In an array, what does an index identify?

### Q3

In a linked list, what does null in a node's next pointer mean?

### Q4

If the first item is at index 4 and its next pointer is 1, where is the second item?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Start pointer — CODE:
int start = 2; — The first logical item is stored at index 2, not necessarily index 0.

## Core idea 2

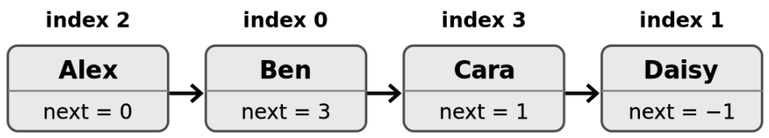
Follow next pointers — CODE:
int current = start;

while (current != -1)
{
Console.WriteLine(data[current]);
current = next[current];
} — Use -1 as null in code. Each pointer decides the next index.

## Core idea 3

Insert at the front — CODE:
next[newIndex] = start;
start = newIndex; — The new item points to the old first item, then start changes.

**Key visual · complete concept**




**start = 2**

**Logical order: Alex → Ben → Cara → Daisy → null**





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 2

Combine, test and explain
Combine traversal, insertion and deletion using the pointer arrays. Record expected and actual outputs for the tests below.
CODE:
Empty list: start=-1
One node: next[start]=-1
Insert into a spare slot
Delete head and middle nodes
Reject insertion if no slot is free
Use -1 only as the null marker. Reuse a removed slot only after marking it free in your chosen allocation scheme.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Identify the first logical item
Run the starter. State the start index and first item. Explain why index 0 is not necessarily the head.
CODE:

```
using System;
public class Program
{
public static void Main()
{
string[] data = {"Ben", "Daisy", "Alex", "Cara", "", ""};
int[] next = {3, -1, 0, 1, -1, -1};
int start = 2; // slots 4 and 5 are unused
Console.WriteLine(data[start]);
}
}
```

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Core idea 4

Remove after current — CODE:
next[current] = next[next[current]]; — The pointer bypasses one node. The array data itself does not need to shift.

## Vocabulary · use the correct technical terms

linked list — A structure where each item contains data and a link to another item.

pointer — A value identifying the location of another item.

start pointer — The index of the first item in the list.

next pointer — The index of the next item.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Traverse Alex → Ben → Cara → Daisy

## Follow the worked method

1. Use the start pointer — CODE:
current = 2
data[2] = "Alex"

2. Continue pointers — CODE:
next[0] = 3
data[3] = "Cara"

next[3] = 1
data[1] = "Daisy"

3. Reach null — CODE:
next[1] = -1
stop

## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Complete the same trace
CODE:
firstIndex = start
secondIndex = ?
secondData = data[?]

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.